

Technology Acceptance and Security Influences on AI Adoption among Small and Medium Enterprises

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Abstract—Small and Medium Enterprises (SMEs) play a vital role in supporting Indonesia's economic growth. However, their adoption of Artificial Intelligence (AI) remains limited despite its potential to enhance efficiency, innovation, and competitiveness. This study extends the Technology Acceptance Model (TAM) to explore how perceived ease of use, perceived usefulness, and perceived security influence SMEs' intention to adopt AI technologies. A quantitative approach was employed using survey data collected from 131 Indonesian SME owners and managers. The data were analyzed using Partial Least Squares-Structural Equation Modeling (PLS-SEM). The findings reveal that perceived ease of use significantly shapes both the perceived usefulness of AI and the intention to adopt it, consistent with TAM's theoretical propositions. Interestingly, perceived usefulness does not affect substantially adoption intention, suggesting that functional benefits alone are insufficient to drive adoption in this context. Instead, perceived security emerges as a critical determinant, indicating that SMEs place strong emphasis on data protection, privacy, and system reliability when considering AI implementation. This study contributes to refining the TAM framework by highlighting the central role of security as a determinant of adoption behavior and demonstrating the limited influence of perceived usefulness in high-risk technological settings. The results underscore the need for policymakers and technology providers to prioritize secure and accessible AI solutions to foster inclusive and sustainable digital transformation among Indonesian SMEs.

Keywords—Artificial Intelligence; Technology Acceptance Model; Security; Perceived Usefulness; Perceived Ease of Use

I. INTRODUCTION

Small and Medium Enterprises (SMEs) are essential to Indonesia's economic growth, as they employ the bulk of the country's workforce and account for more than 60% of its GDP [1]. Artificial Intelligence (AI) has become a key technology that may improve competitiveness, efficiency, and decision-making in the age of rapid digital transformation. Adopting AI is now essential for Indonesian SMEs to prosper in increasingly globalized and dynamic marketplaces, not just a luxury [2].

However, Indonesian SMEs continue to deploy AI technologies unevenly, with several obstacles. Technology acceptability, which includes managers' and entrepreneurs' readiness to incorporate AI technologies into their business operations, is one of the most critical elements affecting adoption [3]. Perceptions of usability, compatibility with current systems, and ease of use influence this adoption. Security is also critical because worries about data privacy,

cyber risks, and regulatory compliance greatly impact trust and the desire to use AI breakthroughs [4].

An unusual setting for studying these processes is Indonesia. Government programs like Making Indonesia 4.0 and the acceleration of the development of the digital economy are driving the nation's rapid increases in internet adoption and digital literacy [5]. SMEs' limitations may hamper the implementation of AI in infrastructure, knowledge, and resources. Therefore, creating strategies that help SMEs leverage AI for sustained growth requires an understanding of how security perceptions and technology acceptability influence AI adoption behavior.

Recent scholarship on AI and technology adoption in SMEs has identified multiple factors shaping successful implementation. Within the Technology–Organization–Environment (TOE) framework [6], drivers such as relative advantage, compatibility, human capital, market demand, and government support emerge as critical, with evidence suggesting that medium-sized firms benefit more strongly from relative advantage than smaller firms [7]. Similarly, research using the Unified Theory of Acceptance and Use of Technology (UTAUT) model shows that effort expectancy and social influence significantly affect behavioral intention to adopt generative AI, with age and work experience moderating these effects [8]. Leadership has also been shown to influence AI adoption through fostering data-driven cultures and enhancing employee competencies, which in turn strengthen supply chain resilience [9]. Yet, in many developing country contexts, SMEs continue to rely primarily on basic information systems rather than advanced technologies such as AI, underscoring the importance of sociocultural and contextualized adoption models [10].

Building upon the preceding discussion, this study offers a novel contribution by examining technology acceptance and security perceptions concurrently as determinants of AI adoption intention among Indonesian SMEs, a sector experiencing rapid digital transformation yet marked by uneven technological uptake. The Technology Acceptance Model (TAM) serves as the central theoretical framework for elucidating user behavior toward emerging technologies, providing a structured lens for understanding the underlying drivers of AI adoption.

In this study, AI technologies refer to practical, business-oriented applications that support SMEs' operational and strategic activities. In Indonesia, SMEs have increasingly adopted various forms of AI, including AI-driven customer

service chatbots, e-commerce recommendation systems, automated inventory and demand forecasting tools, and AI-based digital marketing analytics. For example, small online retailers often use chatbots integrated with WhatsApp or social media platforms to manage customer interactions efficiently. At the same time, food and fashion SMEs rely on AI algorithms for personalized product recommendations and trend forecasting. Furthermore, AI-powered financial management applications assist SMEs in credit scoring, expense tracking, and fraud detection. These examples demonstrate that security concerns, particularly those related to data protection, privacy, and system reliability, are critical factors in determining the extent to which SMEs trust and adopt AI technologies.

In contrast to earlier research that predominantly focuses on organizational, environmental, or leadership dimensions, this study emphasizes three key constructs: perceived ease of use, perceived usefulness, and perceived security. These dimensions capture the practical realities and concerns of SMEs operating in resource-constrained and high-risk environments typical of emerging economies. Through this approach, the research not only advances the existing TAM framework by integrating security as a critical factor but also provides context-specific evidence on how Indonesian SMEs assess the opportunities and challenges of AI implementation. Ultimately, this study seeks to illuminate how perceptions of usability, utility, and security interact to shape adoption behavior, offering actionable insights for policymakers, technology developers, and SME practitioners while contributing to the broader theoretical discourse on technology acceptance and digital transformation.

Despite growing recognition of AI's transformative potential, the mechanisms through which technology acceptance and security shape adoption behavior among Indonesian SMEs remain insufficiently explored. This gap limits policymakers' and practitioners' ability to design effective strategies to address SMEs' specific barriers to digital transformation. Accordingly, this paper is structured as follows: Section II reviews the theoretical foundation and relevant literature; Section III outlines the research methodology; Section IV presents and discusses the empirical findings; and Section V concludes with implications, limitations, and directions for future research.

II. LITERATURE REVIEW

Grounded in the constructs of perceived usefulness and perceived ease of use, the Technology Acceptance Model (TAM) provides a structured framework for understanding how individuals and organizations form intentions to engage with emerging technologies. When integrated with the Theory of Planned Behavior, TAM has demonstrated strong explanatory power, accounting for approximately 59% of the variance in AI acceptance across various contexts [11]. Its adaptability has been evident in multiple domains. For instance, in the hospitality and tourism industry, the development of an interactive TAM (iTAM) highlights the critical role of perceived interactivity in shaping user attitudes toward AI-enabled service robots [12]. Similarly, studies combining TAM with status quo bias theory in the context of AI voice assistants show that perceived ease of use and perceived value positively influence user attitudes, whereas switching costs and perceived risks act as barriers to adoption [13]. In the education sector, TAM has been employed to examine teachers' intentions to utilize AI-based e-learning

platforms, revealing that perceptions of usefulness and emotional engagement significantly impact adoption decisions [14]. Collectively, these findings underscore TAM's relevance as a theoretical lens for exploring how perceptions of utility, usability, and security interact to shape AI adoption behaviors—particularly in dynamic, resource-constrained environments such as SMEs.

Studies investigating the impact of Perceived Ease of Use (PEU) on AI adoption intentions (IAI) demonstrate consistently favorable correlations across several areas. In accounting education, PEU serves as a mediator between technological preparedness and aspirations to use AI, in conjunction with perceived utility [15]. Likewise, PEU has substantial positive effects on both the desire to use and the actual adoption of digital technologies, such as e-wallets [16]. In the realm of AI-enabled smart speakers, PEU is a pivotal aspect impacting attitudes and intentions for continued usage, whereas social elements such as media richness and parasocial interactions shape perceptions of PEU [17]. These results repeatedly validate PEU as an essential determinant in AI adoption across different contexts and user demographics.

[18] discovered that in small and medium firms in Taiwan, only perceived ease of use (PEU) had a direct impact on perceived usefulness (PU), and PU was the only direct predictor of system utilization. This is different from the original TAM research by [19], which revealed that PU had a significant effect on user intentions, whereas PEU had a weaker but still significant effect that weakened over time. The positive correlation between PEU and PU has been consistently validated across several areas. [20] validated that PEU substantially influences PU in the context of e-book uptake among university students. [15] also discovered that both PEU and PU act as intermediaries in the interaction between technological readiness and the adoption of artificial intelligence in accounting contexts, demonstrating their ongoing significance in contemporary technology acceptance frameworks.

PU acts as a bridge between students' readiness to use technology and their plans to use AI [15]. In the same way, studies of employees demonstrate that PU completely mediates the negative link between job instability and plans to use AI, and somewhat mediates the positive link between habit and plans to utilize AI at work [21]. PU affects more than only AI-specific applications. For example, research on the adoption of e-wallets demonstrates that PU has a significant beneficial impact on both usage intentions and actual adoption behaviors [16]. In AI assistant-enabled smart speakers, perceived usefulness (PU) is a significant factor that affects how users feel about the technology and whether they plan to keep using it. It works with the perceived simplicity of use to encourage people to adopt the technology [22]. These results consistently affirm PU as a critical factor in technology adoption decisions.

In the realm of financial services, technical optimism enhances the intention to utilize AI-driven robo-advisors, but insecurity diminishes adoption intentions; conversely, technological discomfort unexpectedly had good outcomes [23]. In terms of security issues, insecurity makes people think that AI systems are not very beneficial, especially in retail settings, where being inventive and optimistic makes both PEOU and PU go up [24]. These findings indicate that security concerns and feelings of unease consistently act as obstacles to AI adoption across diverse fields.

III. METHODS

This study employed a quantitative research design to investigate the determinants of AI adoption among small and medium enterprises (SMEs) in Indonesia. A structured questionnaire consisting of closed-ended items was developed to measure respondents' perceptions of the key constructs under investigation. Data were collected using a cross-sectional design, capturing responses at a single point in time to reflect the current state of AI awareness and adoption-readiness among Indonesian SMEs. The research specifically targeted business owners and managers who possessed at least a basic understanding of AI applications relevant to their operational or strategic activities.

The data collection process was conducted through an online survey distributed via social media channels, including Instagram and WhatsApp. These platforms were selected to ensure a wide reach and accessibility for SMEs located across diverse regions. Questionnaires were chosen as the primary instrument because they are more cost-effective and time-efficient compared to interviews or observations [25]. Participation was voluntary, and a purposive sampling strategy was used to ensure respondents were directly involved in decision-making within their respective organizations. This approach enhanced the relevance and accuracy of the collected data, aligning with the study's objective of exploring managerial perspectives on AI adoption. The dataset for this article is available at [26].

The adequacy of the sample size was determined following recommendations for Partial Least Squares–Structural Equation Modeling (PLS-SEM). In accordance with the “ten-times rule” proposed by [27], the minimum required sample should be at least ten times the maximum number of structural paths directed toward a latent construct in the model. Since the most complex construct in this study had three incoming paths, a minimum of 30 participants was deemed sufficient. The final dataset consisted of 131 valid responses, surpassing the minimum requirement and providing a robust basis for model estimation and hypothesis testing.

The conceptual framework for this study was grounded in the Technology Acceptance Model (TAM), which was extended by incorporating perceived security as an additional determinant of AI adoption intention. The model included four key constructs: perceived ease of use, perceived usefulness, perceived security, and behavioral intention to adopt AI. The research process followed three main stages: instrument development, data collection, and model analysis. During the first stage, measurement items were adapted from established TAM literature and refined to suit the SME context. The second stage involved data screening and validation to ensure completeness and consistency. The final stage used PLS-SEM to assess the measurement model's reliability and validity, followed by testing the hypothesized relationships among the constructs in the structural model.

This methodological design provides a systematic, empirically grounded approach to examining how perceptions of usability, utility, and security jointly influence AI adoption behavior among Indonesian SMEs. By adopting the TAM framework and applying PLS-SEM analysis [28], the study ensures both theoretical rigor and analytical robustness in capturing the complex dynamics underlying technology acceptance in emerging market contexts.

IV. RESULTS AND DISCUSSIONS

The respondents' demographic profile offers crucial background information for comprehending how SMEs are using AI. Female respondents made up the majority (51.7%), followed by male respondents (47.7%), with 0.6 percent choosing not to reveal their gender. Given the minor preponderance of female entrepreneurs or SME managers in the sample, this distribution suggests that involvement was fairly balanced between genders.

SME leaders and decision-makers are primarily in their middle working years, as evidenced by the fact that the majority of respondents were in the productive age groups of 30–39 (32.5%) and 40–49 (31.1%). Younger people (20–29 years, 21.2%) and older people (50–59 years, 12.6%) accounted for a smaller share. Just 0.8% of respondents were under 20 years old, while 1.8% were over 60. This suggests that adults in Indonesia hold the majority of SME management and ownership positions, with established work histories.

Regarding educational background, the majority of respondents had completed secondary school or its equivalent (43.7%) and university programs (47%). A lower percentage had earned postgraduate degrees, such as master's (7.9%) and doctoral (1.4%). According to this distribution, secondary education remains the basis for business operations, despite a sizable share of SME actors holding advanced degrees.

The majority of SMEs surveyed had been in business for one to five years (58.8%), with those with six to ten years of operation coming in second (18.3%). Only a small percentage of enterprises had been in operation for more than ten years (less than 6% combined), while a smaller percentage were relatively young, having been in operation for less than one year (16.8%). In line with the broader growth of the digital economy, this trend shows the increasing presence of relatively young SMEs in Indonesia's business environment.

Based on Table I, all indicators show outer loadings above the recommended threshold of 0.70, indicating good indicator reliability [28]. All constructs exceed the recommended threshold of 0.70, indicating strong internal consistency. All AVE values exceed the recommended threshold of 0.50, indicating good convergent validity.

TABLE I. OUTER LOADINGS, COMPOSITE RELIABILITY, AND AVERAGE VARIANCE EXTRACTED (AVE)

	Original Sample (O)	Standard Deviation (STDEV)	p Values	CR and AVE
IAI1 <- IAI	0,836	0,036	0,000	CR = 0,921 AVE = 0,745
IAI2 <- IAI	0,901	0,027	0,000	
IAI3 <- IAI	0,900	0,021	0,000	
IAI4 <- IAI	0,810	0,043	0,000	
PEU1 <- PEU	0,897	0,019	0,000	CR = 0,916 AVE = 0,784
PEU2 <- PEU	0,885	0,037	0,000	
PEU3 <- PEU	0,873	0,054	0,000	
PU1 <- PU	0,886	0,043	0,000	CR = 0,920 AVE = 0,792
PU2 <- PU	0,872	0,052	0,000	
PU3 <- PU	0,913	0,026	0,000	
SEC1 <- SEC	0,889	0,028	0,000	CR = 0,879 AVE = 0,712
SEC2 <- SEC	0,687	0,083	0,000	
SEC3 <- SEC	0,935	0,012	0,000	

The diagonal values (bold, e.g., 0.863 for IAI, 0.885 for PEU, 0.890 for PU, 0.844 for SEC) represent the square root of AVE for each construct (see Table II). For discriminant

validity, each diagonal value should be greater than the correlations with other constructs in the same row/column. Strong internal consistency across all constructs, as demonstrated by Cronbach's Alpha and rho_A values (>0.70).

TABLE II. DISCRIMINANT VALIDITY AND RELIABILITY TEST

	IAI	PEU	PU	SEC	Cronbach's Alpha	rho_A
IAI	0.863				0.886	0.897
PEU	0.434	0.885			0.862	0.868
PU	0.482	0.366	0.890		0.870	0.895
SEC	0.575	0.425	0.734	0.844	0.790	0.822

Based on Table III, Perceived Ease of Use significantly influences AI Adoption Intention. Although the effect size is small, this confirms that ease of use plays a role in shaping adoption intention among SMEs. Perceived Ease of Use significantly affects Perceived Usefulness. This aligns with TAM theory, which suggests that ease of use enhances perceptions of usefulness. Perceived Usefulness does not significantly influence AI Adoption Intention. Security has a significant and relatively strong effect on AI Adoption Intention. Concerns about data protection and system trustworthiness play a crucial role in adoption decisions.

TABLE III. HYPOTHESES TESTS

	Original Sample (O)	Standard Deviation (STDEV)	P Values	f Square	Note
PEU → IAI	0,224	0,080	0,006	0,066	H1 accepted
PEU → PU	0,366	0,083	0,000	0,155	H2 accepted
PU → IAI	0,105	0,114	0,359	0,008	H3 rejected
SEC → IAI	0,403	0,092	0,000	0,113	H4 accepted

The discovery that PEU substantially affects the intention to IAI (H1 accepted) corroborates the Technology Acceptance Model's (TAM) assertion that usability perceptions are a pivotal factor in adoption behavior. Furthermore, the robust positive correlation between PEU and PU (H2 accepted) corroborates the recognized mediating role of usefulness, in which ease of use amplifies perceptions of utility. This study aligns with the TAM and underscores the need to develop AI solutions that are both useful and intuitive for SMEs that may lack extensive technical skills.

Interestingly, PU did not substantially affect the IAI (H3 rejected). This contradicts a fundamental premise of TAM, indicating that, in the realm of AI adoption among Indonesian SMEs, mere usefulness is insufficient to influence behavioral intention. A reasonable argument is that SMEs acknowledge the potential benefits of AI but are reluctant to implement it owing to perceived risks, limited resources, or insufficient preparedness. This research enhances theoretical understanding by highlighting the limitations of the TAM and suggesting that usefulness may be subordinate to other considerations, such as security and usability, in circumstances characterized by significant technological uncertainty.

The study shows that the SEC has a meaningful influence on the IAI (H4 accepted). This expands the TAM framework by incorporating a critical contextual element—security—into

the adoption model. The substantial effect size ($f^2 = 0.113$) underscores the importance of security as both an external factor and a fundamental predictor of adoption intention in AI contexts.

Lower PLS prediction errors when compared to the LM benchmark and positive Q²_predict values, which collectively show medium predictive power, show that the model predicts AI Adoption Intention (IAI) indicators quite effectively [29] (see Table IV).

TABLE IV. PLS PREDICT

Indicator	Comparison (PLS vs. LM Errors)	Q ² _predict	Interpretation of Predictive Power
Construct: IAI (AI Adoption Intention)			
IAI1	PLS errors < LM (1.132 vs. 1.152 RMSE)	0.300	Medium predictive power
IAI2	PLS errors < LM (1.104 vs. 1.117 RMSE)	0.279	Medium predictive power
IAI4	PLS errors < LM (1.170 vs. 1.187 RMSE)	0.152	Medium predictive power
IAI3	PLS errors < LM (1.062 vs. 1.087 RMSE)	0.270	Medium predictive power
Construct: PU (Perceived Usefulness)			
PU3	LM errors < PLS (0.842 vs. 1.032 RMSE)	0.130	Weak predictive power
PU2	LM errors < PLS (0.917 vs. 1.073 RMSE)	0.039	Weak predictive power
PU1	LM errors < PLS (0.876 vs. 1.084 RMSE)	0.087	Weak predictive power

As shown in Fig. 1 (SEM Model), the structural relationships confirm that ease of use and security are the strongest predictors of AI adoption intention.

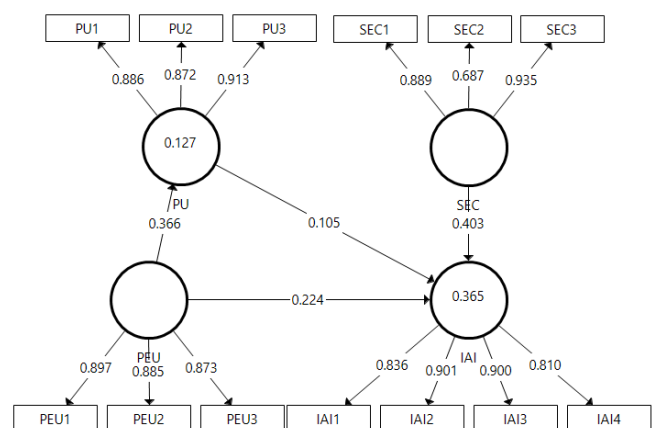


Fig. 1. SEM Model

V. CONCLUSIONS

The findings of this study offer several theoretical contributions to the literature on technology acceptability and AI adoption in SMEs. This research advances theoretical development by demonstrating that, for nascent and data-

intensive technologies such as AI, perceptions of security may supersede perceptions of utility in shaping adoption decisions, especially in resource-limited contexts such as those of SMEs in Indonesia.

This study underscores the significance of security and user-friendliness in influencing Indonesian SMEs' propensity to adopt AI. These findings indicate numerous beneficial recommendations that can be implemented. To facilitate the safe and secure use of AI by small and medium-sized businesses (SMEs), policymakers and government agencies should enhance the effectiveness of their awareness campaigns and increase the availability of training. Technology companies must offer AI solutions that are user-friendly, require minimal technical expertise, and are protected by robust security measures to establish trust with SMEs. SME proprietors and managers may also consider gradually integrating AI into their operations through secure, scalable platforms. This approach would mitigate perceived risks while still leveraging the technology's benefits.

Additional research is recommended in numerous directions. Initially, a more comprehensive understanding of adoption behavior may be achieved by incorporating additional relevant factors, such as trust, cost, or organizational preparedness, into the model. Secondly, further research could employ a longitudinal approach to investigate the development of attitudes toward security, ease of use, and utility as small and medium-sized enterprises build their AI experience. Third, comparative studies of SMEs in Indonesia and other emerging economies may reveal cultural or institutional disparities that influence AI adoption. Lastly, qualitative methods such as interviews or case studies may complement survey results by providing additional insight into the reasons and challenges SMEs encounter when implementing a new technology.

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AUTHOR CONTRIBUTORSHIP

Yuniarty and Hartiwi worked closely together in shaping the overall research concept and design. They were responsible for data collection, formal analysis, funding acquisition, and project administration, as well as managing research resources and validating the study's findings. Both also contributed to the visualization of results and the preparation of the initial manuscript draft. Ridho led the development of the research methodology and managed the analytical tools used in the study, while also providing supervision and assisting in the validation and drafting process. Noe contributed primarily to the review and refinement of the manuscript, ensuring that the final version was clear, coherent, and met scholarly publication standards. All authors discussed the results, contributed to the final manuscript, and approved it for submission.

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